

the cancerous cells and that these undetected malignant cells have started to multiply out of control.) It is not feasible, in the present state of technology, to examine every single one of the patient's cells to see if all of them are nonmalignant, so the doctor takes a sample by scanning the body with as much precision as possible. Then she makes an assumption about what she did not see. I was once taken aback when a doctor told me after a routine cancer checkup, "Stop worrying, we have evidence of cure." "Why?" I asked. "There is evidence of *no* cancer" was the reply. "How do you know?" I asked. He replied, "The scan is negative." Yet he went around calling himself doctor!

An acronym used in the medical literature is NED, which stands for No Evidence of Disease. There is no such thing as END, Evidence of No Disease. Yet my experience discussing this matter with plenty of doctors, even those who publish papers on their results, is that many slip into the round-trip fallacy during conversation.

Doctors in the midst of the scientific arrogance of the 1960s looked down at mothers' milk as something primitive, as if it could be replicated by their laboratories—not realizing that mothers' milk might include useful components that could have eluded their scientific understanding—a simple confusion of *absence of evidence* of the benefits of mothers' milk with *evidence of absence* of the benefits (another case of Platonicity as "it did not make sense" to breast-feed when we could simply use bottles). Many people paid the price for this naïve inference: those who were not breast-fed as infants turned out to be at an increased risk of a collection of health problems, including a higher likelihood of developing certain types of cancer—there had to be in mothers' milk some necessary nutrients that still elude us. Furthermore, benefits to mothers who breast-feed were also neglected, such as a reduction in the risk of breast cancer.

Likewise with tonsils: the removal of tonsils may lead to a higher incidence of throat cancer, but for decades doctors never suspected that this "useless" tissue might actually have a use that escaped their detection. The same with the dietary fiber found in fruits and vegetables: doctors in the 1960s found it useless because they saw no immediate evidence of its necessity, and so they created a malnourished generation. Fiber, it turns out, acts to slow down the absorption of sugars in the blood and scrapes the intestinal tract of precancerous cells. Indeed medicine has caused plenty of damage throughout history, owing to this simple kind of inferential confusion.

I am not saying here that doctors should not have beliefs, only that some kinds of definitive, closed beliefs need to be avoided—this is what Menodotus and his school seemed to be advocating with their brand of skeptical-empirical medicine that avoided theorizing. Medicine has gotten better—but many kinds of knowledge have not.